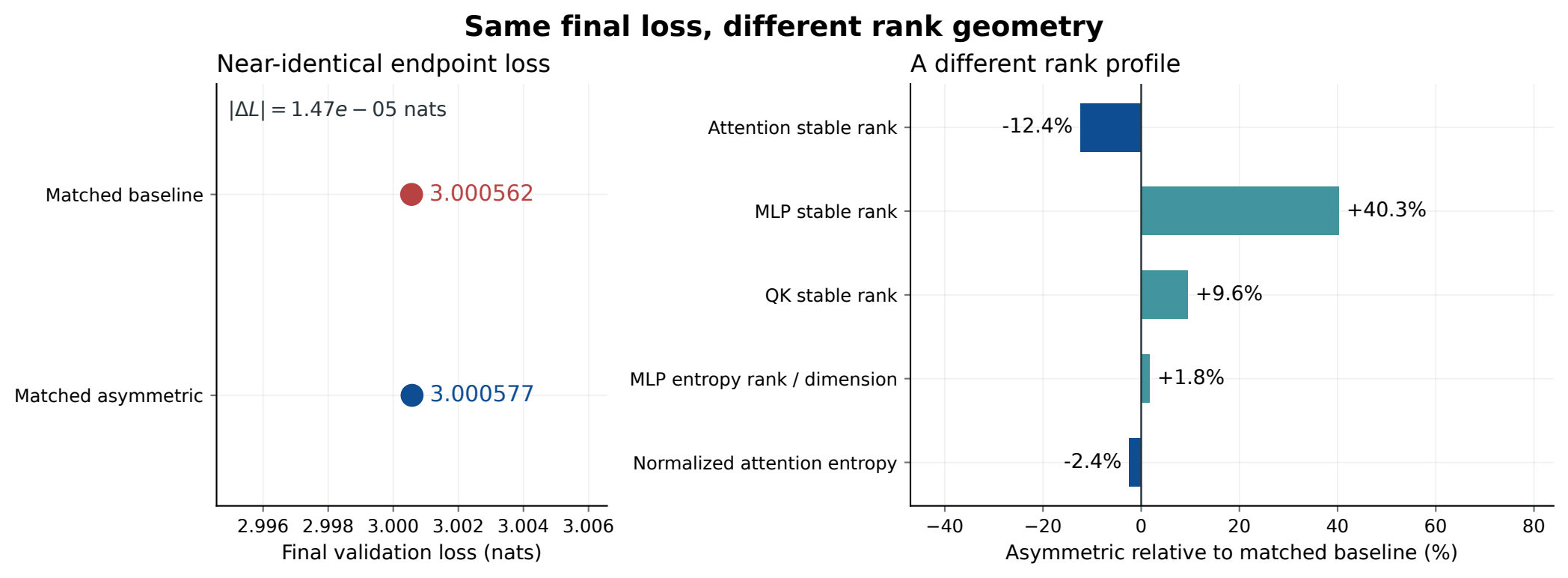
Same final loss, different rank geometry
Near-identical endpoint loss
$|\Delta L| = 1.47e-05$ nats
Matched baseline
3.000562
Matched asymmetric
3.000577
2.996
2.998
3.000
3.002
3.004
3.006
Final validation loss (nats)
A different rank profile
Attention stable rank
-12.4%
MLP stable rank
+40.3%
QK stable rank
+9.6%
MLP entropy rank / dimension
+1.8%
Normalized attention entropy
-2.4%
−40
−20
0
20
40
60
80
Asymmetric relative to matched baseline (%)